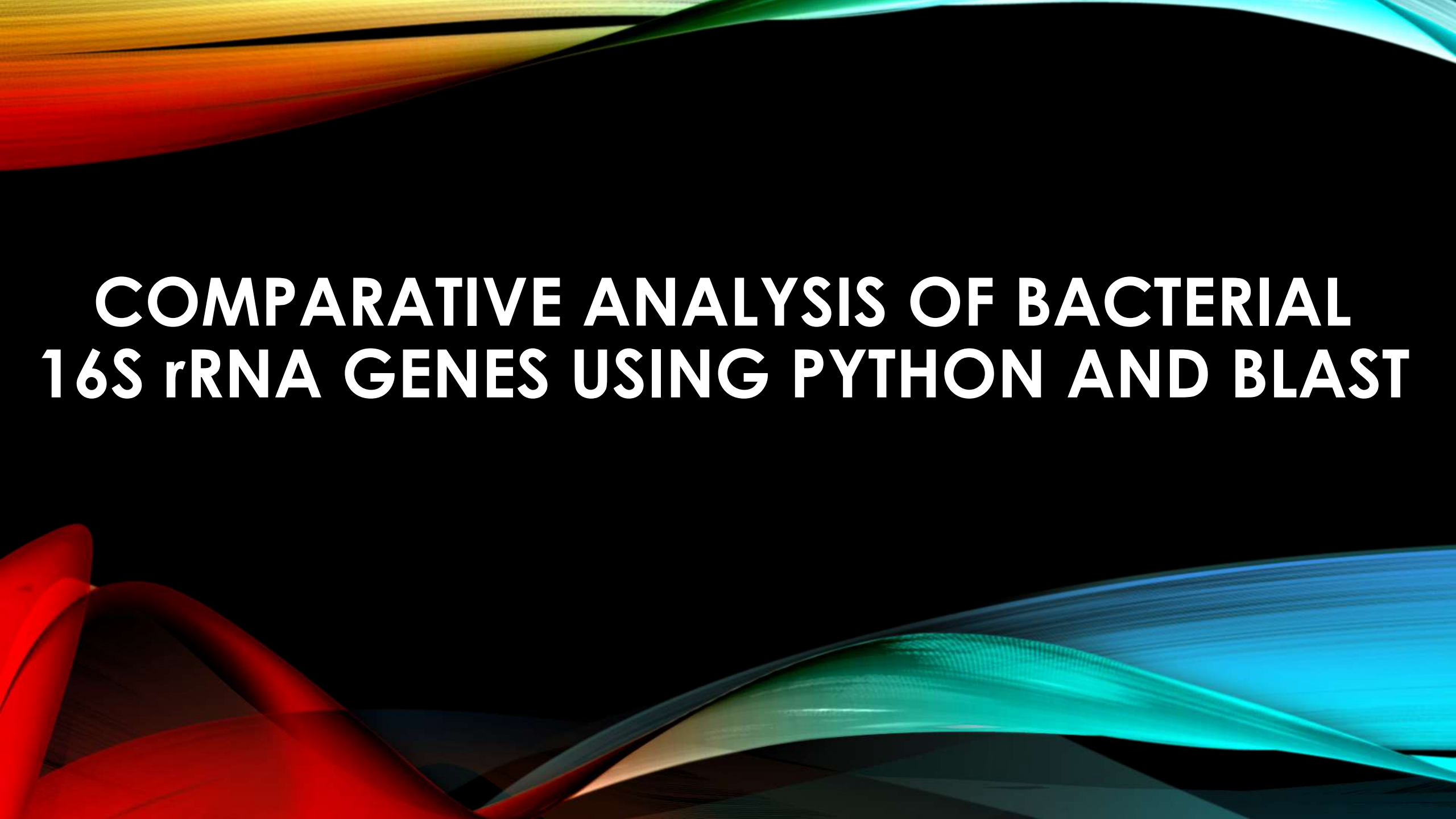




COMPARATIVE ANALYSIS OF BACTERIAL 16S rRNA GENES USING PYTHON AND BLAST



PROJECT OVERVIEW

- The project performs a comparative bioinformatics analysis of bacterial 16S rRNA gene sequences.
- Python and Biopython were used to extract biological features such as sequence length and GC content, followed by BLASTn analysis to validate taxonomic identity.



OBJECTIVES

- Analysing bacterial 16S rRNA sequences.
- Extracted sequence length and GC content/fraction using Python.
- Validating species identity using BLASTn.
- Interpreting biological significance of results.
- Visualized comparative patterns across organisms.



TOOLS & TECHNOLOGIES USED

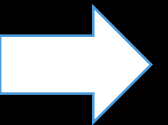
- Python
- Biopython
- Pandas
- Matplotlib
- NCBI BLAST
- Google Colab



DATASET DESCRIPTION

- Bacterial 16S rRNA FASTA sequences downloaded from NCBI
- Organisms include *Klebsiella*, *Escherichia coli*, *Salmonella*, *Pseudomonas* and *Staphylococcus aureus*.

METHODOLOGY



A. DOWNLOADED 16S rRNA SEQUENCES FROM NCBI

NIH National Library of Medicine
National Center for Biotechnology Information

Nucleotide: Nucleotide

GenBank: Send to:

Klebsiella pneumoniae CC17 gene for 16S rRNA, partial sequence
GenBank: LC668253.1
FASTA [GenBank](#)

LOCUS: LC668253.1 1410 bp DNA linear ECT 06-NOV-2025
DEFINITION: Klebsiella pneumoniae CC17 gene for 16S rRNA, partial sequence.
ACCESSION: LC668253.1
VERSION: LC668253.1
KEYWORDS: -
SOURCE: Klebsiella pneumoniae
ORGANISM: Klebsiella pneumoniae
Bacteria; Pseudomonadota; Pseudomonadota; Gammaproteobacteria;
Enterobacteriales; Enterobacteriaceae; Klebsiella/Pseudomonas group;
Klebsiella; Klebsiella pneumoniae complex;
REFERENCE: 1
AUTHORS: Chanchao, C., Eslamloo, G. and Ita, T.
TITLE: Partial 16S rRNA gene sequence of bacteria isolated from dehydrated
sludge
JOURNAL: Unpublished
REFERENCE: 2 (bases 1 to 1410)
AUTHORS: Eslamloo, G., Chanchao, C. and Ita, T.
TITLE: Direct Subcloning
JOURNAL: Submitted (18-APR-2025) Contact:Frederic The Serna University;

Analyze this sequence
Run BLAST
Pick Primers
Highlight Sequence Features
Find in this Sequence
Related information
Taxonomy
Recent activity
LinkOut to external resources
3d/4d SDS Database (TCW)

NIH National Library of Medicine
National Center for Biotechnology Information

Nucleotide: Nucleotide

GenBank: Send to:

Salmonella enterica subsp. enterica strain 6 gene for 16S rRNA, partial sequence
GenBank: LC760081.1
FASTA [GenBank](#)

LOCUS: LC760081.1 1413 bp DNA linear ECT 19-NOV-2025
DEFINITION: Salmonella enterica subsp. enterica strain 6 gene for 16S rRNA,
partial sequence.
ACCESSION: LC760081.1
VERSION: LC760081.1
KEYWORDS: -
SOURCE: Salmonella enterica subsp. enterica
ORGANISM: Salmonella enterica subsp. enterica
Bacteria; Pseudomonadota; Pseudomonadota; Gammaproteobacteria;
Enterobacteriales; Enterobacteriaceae; Salmonella.
REFERENCE: 1
AUTHORS: Lv, H. and Wang, Z.
TITLE: Prevention and control of microbial contamination of chicken powder
seasoning
JOURNAL: Unpublished
REFERENCE: 2 (bases 1 to 1413)
AUTHORS: Lv, H. and Wang, Z.
TITLE: Direct Subcloning
JOURNAL: Submitted (08-NOV-2025) Contact:Shao Lv Hean University of

Analyze this sequence
Run BLAST
Pick Primers
Highlight Sequence Features
Find in this Sequence
Related information
Taxonomy
Recent activity
LinkOut to external resources
3d/4d SDS Database (TCW)

NIH National Library of Medicine
National Center for Biotechnology Information

Nucleotide: Nucleotide

GenBank: Send to:

Pseudomonas fulva 67 16S ribosomal RNA gene, partial sequence
GenBank: F397259.1
FASTA [GenBank](#)

LOCUS: F397259.1 1413 bp DNA linear ECT 06-NOV-2025
DEFINITION: Pseudomonas fulva 67 16S ribosomal RNA gene, partial sequence.
ACCESSION: F397259.1
VERSION: F397259.1
KEYWORDS: -
SOURCE: Pseudomonas fulva
ORGANISM: Pseudomonas fulva
Bacteria; Pseudomonadota; Pseudomonadota; Gammaproteobacteria;
Pseudomonadales; Pseudomonadaceae; Pseudomonas.
REFERENCE: 1 (bases 1 to 1413)
AUTHORS: Wu, L.-L., Ou, F.-T., He, S.-R., Wang, J.-Y., Chai, Y.-H., Han, J.-Y.,
Luo, B.-Y., Chai, G.-H., Liu, J.-K. and Kang, D.-S.
TITLE: Isolation and characterization of 1411 bacteria in Guangxi City area
JOURNAL: Unpublished
REFERENCE: 2 (bases 1 to 1413)
AUTHORS: Kang, D.-S. and Chai, Y.-T.
TITLE: Direct Subcloning
JOURNAL: Submitted (28-APR-2009) Molecular Genetics laboratory, School of
Biological Sciences and Technology, Chongqing National University;

Analyze this sequence
Run BLAST
Pick Primers
Highlight Sequence Features
Find in this Sequence
Related information
Taxonomy
LinkOut to external resources
3d/4d SDS Database (TCW)

NIH National Library of Medicine
National Center for Biotechnology Information

Nucleotide: Nucleotide

GenBank: Send to:

Escherichia coli Gost-EC-008 gene for 16S rRNA, partial sequence
GenBank: LC070801.1
FASTA [GenBank](#)

LOCUS: LC070801.1 1410 bp DNA linear ECT 18-NOV-2025
DEFINITION: Escherichia coli Gost-EC-008 gene for 16S rRNA, partial sequence.
ACCESSION: LC070801.1
VERSION: LC070801.1
KEYWORDS: -
SOURCE: Escherichia coli
ORGANISM: Escherichia coli
Bacteria; Pseudomonadota; Pseudomonadota; Gammaproteobacteria;
Enterobacteriales; Enterobacteriaceae; Escherichia.
REFERENCE: 1
AUTHORS: Vatsani, A. E.
TITLE: Escherichia coli 16S ribosomal RNA gene, partial sequence
JOURNAL: Unpublished
REFERENCE: 2 (bases 1 to 1410)
AUTHORS: Vatsani, A. E. and Bhat, A.
TITLE: Direct Subcloning
JOURNAL: Submitted (14-NOV-2005) Contact:Shao Hean University of
Agriculture and Technology, Botany-Biological
Sciences, Nairobi South 4, Diamond Park 1, Nairobi, Nairobi

Analyze this sequence
Run BLAST
Pick Primers
Highlight Sequence Features
Find in this Sequence
Related information
Taxonomy
Recent activity
LinkOut to external resources
3d/4d SDS Database (TCW)

NIH National Library of Medicine
National Center for Biotechnology Information

Nucleotide: Nucleotide

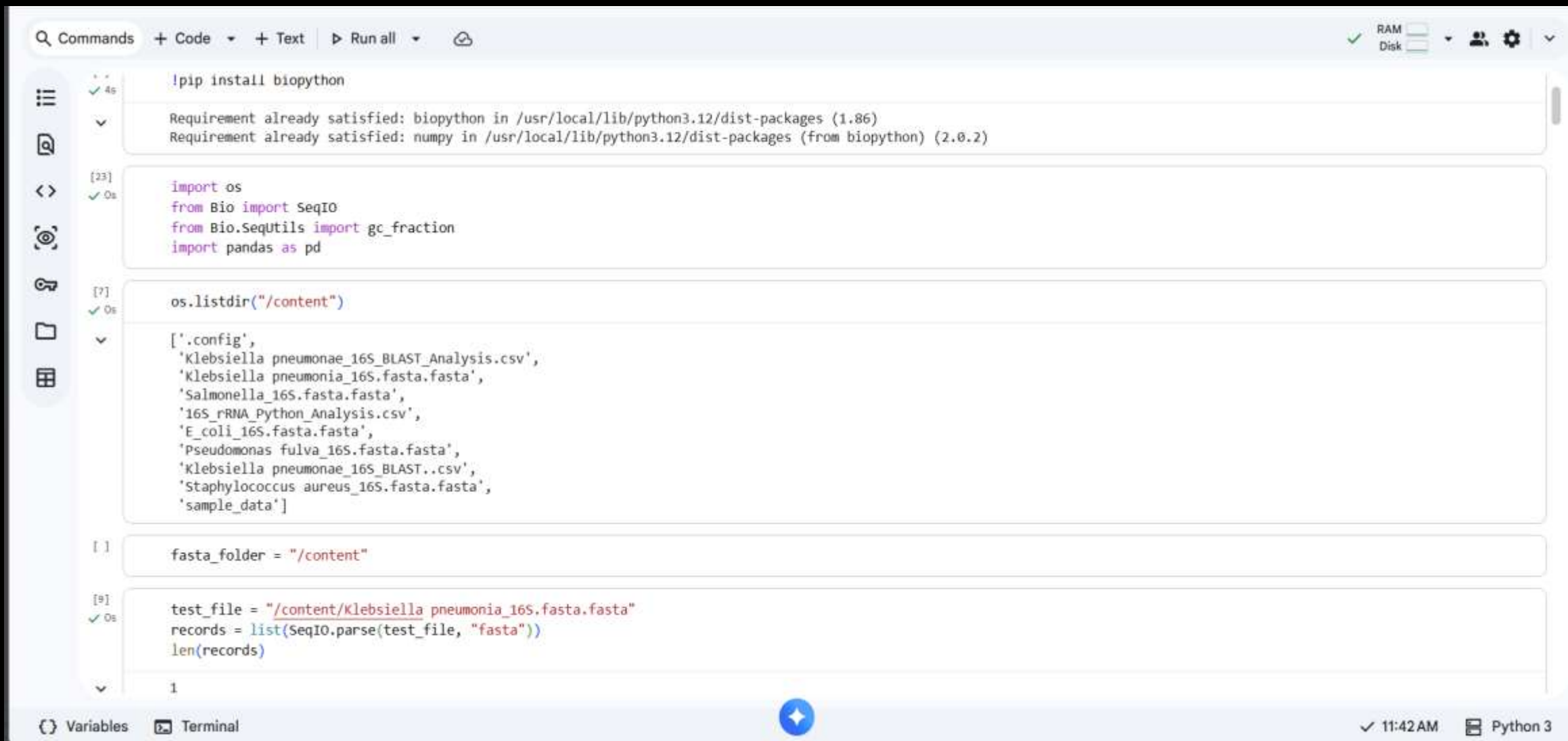
GenBank: Send to:

Staphylococcus aureus Iraq'YaAwSh22.10 gene for 16S rRNA, partial sequence
GenBank: LC291138.1
FASTA [GenBank](#)

LOCUS: LC291138.1 1424 bp DNA linear ECT 14-SEP-2005
DEFINITION: Staphylococcus aureus Iraq'YaAwSh22.10 gene for 16S rRNA, partial
sequence.
ACCESSION: LC291138.1
VERSION: LC291138.1
KEYWORDS: -
SOURCE: Staphylococcus aureus
ORGANISM: Staphylococcus aureus
Bacteria; Bacillales; Bacillales; Bacilli; Bacillales;
Staphylococcales; Staphylococcus.
REFERENCE: 1
AUTHORS: Kadhe, S. K., Yousif, F. S. and Jang, A. H.
TITLE: 16S rRNA gene in Staphylococcus aureus
JOURNAL: Unpublished
REFERENCE: 2 (bases 1 to 1424)
AUTHORS: Kadhe, S. K., Yousif, F. S. and Jang, A. H.
TITLE: Direct Subcloning
JOURNAL: Submitted (23-SEP-2005) Contact:Shao Hean University of
Biology, Nairobi, Nairobi 00001, Nairobi 00001

Analyze this sequence
Run BLAST
Pick Primers
Highlight Sequence Features
Find in this Sequence
Related information
Taxonomy
Recent activity
LinkOut to external resources
3d/4d SDS Database (TCW)

B. PARSED FASTA FILES USING BIOPYTHON



The screenshot displays a Jupyter Notebook environment with a sidebar on the left containing icons for file management, search, and other functions. The top bar includes a search field, tabs for 'Commands', 'Code', and 'Text', and a 'Run all' button. The main area shows a series of code cells with their execution status and output.

```
!pip install biopython
```

Requirement already satisfied: biopython in /usr/local/lib/python3.12/dist-packages (1.86)
Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages (from biopython) (2.0.2)

```
[23]: import os
      from Bio import SeqIO
      from Bio.Sequtils import gc_fraction
      import pandas as pd
```

```
[7]: os.listdir("/content")
```

```
[ ]: ['.config',
      'Klebsiella pneumoniae_16S_BLAST_Analysis.csv',
      'Klebsiella pneumonia_16S.fasta.fasta',
      'Salmonella_16S.fasta.fasta',
      '16S_rRNA_Python_Analysis.csv',
      'E_coli_16S.fasta.fasta',
      'Pseudomonas fulva_16S.fasta.fasta',
      'Klebsiella pneumoniae_16S_BLAST..csv',
      'Staphylococcus aureus_16S.fasta.fasta',
      'sample_data']
```

```
[ ]: fasta_folder = "/content"
```

```
[9]: test_file = "/content/Klebsiella pneumonia_16S.fasta.fasta"
      records = list(SeqIO.parse(test_file, "fasta"))
      len(records)
```

1

At the bottom, there are tabs for 'Variables' and 'Terminal', a blue circular button with a white plus sign, and a status bar showing '✓ 11:42 AM' and 'Python 3'.

C. EXTRACTED SEQUENCE LENGTH AND GC-FRACTION

```
Q Commands + Code + Text ▶ Run all +
[14] ✓ Os
data = []
for file_name in os.listdir("/content"):
    if file_name.endswith(".fasta"):
        file_path = os.path.join("/content", file_name)
        for record in SeqIO.parse(file_path, "fasta"):
            organism = file_name.replace("_16S.fasta.fasta", "").replace("_", " ")
            sequence = record.seq

            length = len(sequence)
            gc_content = gc_fraction(sequence) * 100

            data.append([organism, length, gc_content])

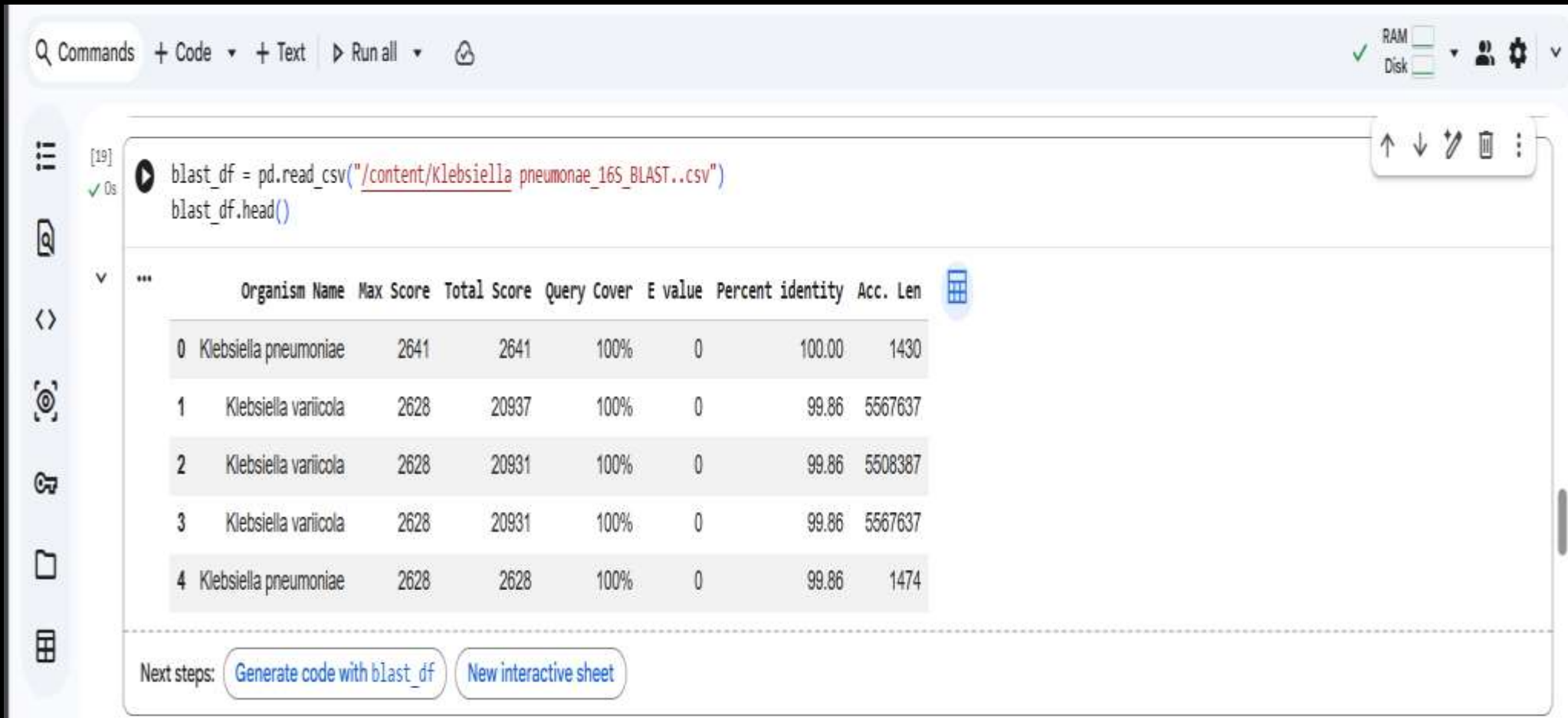
df = pd.DataFrame(
    data,
    columns=["Organism", "Sequence_Length", "gc_fraction (%)"]
)

df
```

	Organism	Sequence_Length	gc_fraction (%)
0	Klebsiella pneumonia	1430	55.244755
1	Salmonella	1413	54.706299
2	E coli	1086	55.064457
3	Pseudomonas fulva	1415	53.780919
4	Staphylococcus aureus	1424	51.053371

D.PERFORMED BLASTN ANALYSIS FOR TAXONOMIC VALIDATION & ANALYSED BLAST OUTPUTS USING PANDAS

1)



The screenshot displays a Jupyter Notebook environment. The top toolbar includes a search icon, a 'Commands' dropdown, and buttons for '+ Code', '+ Text', 'Run all', and a refresh icon. On the right, there are status indicators for RAM and Disk, along with user and settings icons. The left sidebar contains navigation icons for a table of contents, search, code editor, and other notebook functions.

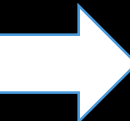
The main area shows a code cell with the following Python code:

```
[19]: blast_df = pd.read_csv("/content/Klebsiella pneumoniae_16S_BLAST..csv")
      blast_df.head()
```

The output of the code is a pandas DataFrame with the following columns: Organism Name, Max Score, Total Score, Query Cover, E value, Percent identity, and Acc. Len. The DataFrame contains 5 rows of data.

	Organism Name	Max Score	Total Score	Query Cover	E value	Percent identity	Acc. Len
0	Klebsiella pneumoniae	2641	2641	100%	0	100.00	1430
1	Klebsiella varicola	2628	20937	100%	0	99.86	5567637
2	Klebsiella varicola	2628	20931	100%	0	99.86	5508387
3	Klebsiella varicola	2628	20931	100%	0	99.86	5567637
4	Klebsiella pneumoniae	2628	2628	100%	0	99.86	1474

At the bottom of the cell, there are two buttons: 'Generate code with blast_df' and 'New interactive sheet'.



2)

Q Commands + Code + Text ▶ Run all

✓ RAM Disk

[28]
✓ 0s

```
blast_clean = blast_df[
    ["Organism Name", "Query Cover", "E value", "Percent identity"]
]
blast_clean.head()
```

	Organism Name	Query Cover	E value	Percent identity
0	Klebsiella pneumoniae	100%	0	100.00
1	Klebsiella variicola	100%	0	99.86
2	Klebsiella variicola	100%	0	99.86
3	Klebsiella variicola	100%	0	99.86
4	Klebsiella pneumoniae	100%	0	99.86

Next steps: [Generate code with blast_clean](#) [New interactive sheet](#)



3)

☰

🔍

⏏

🎯

🔑

📁

📊

[21]
✓ 0s

▶

```
high_confidence = blast_clean[
    (blast_clean["Percent identity"] >= 97) &
    (blast_clean["E value"] == 0)
]
high_confidence.head()
```

▼

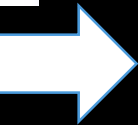
...

	Organism Name	Query Cover	E value	Percent identity
0	Klebsiella pneumoniae	100%	0	100.00
1	Klebsiella variicola	100%	0	99.86
2	Klebsiella variicola	100%	0	99.86
3	Klebsiella variicola	100%	0	99.86
4	Klebsiella pneumoniae	100%	0	99.86

Next steps:

[Generate code with high_confidence](#)

[New interactive sheet](#)



INTERPRETED BIOLOGICAL SIGNIFICANCE OF RESULTS

Interpreting 16S rRNA Sequence Length

- Sequence lengths ≈ **1400–1600 bp** and very similar across organisms
- 16S rRNA is a **highly conserved gene**, Conservation = reliable taxonomic marker
- Therefore, the conserved length of the 16S rRNA gene across organisms confirms its suitability for bacterial identification and phylogenetic analysis

Interpreting GC Fraction Differences

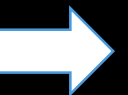
- GC content reflects: Genomic composition, Evolutionary lineage & Environmental adaptation
- GC % varies between organisms : *Pseudomonas* → higher GC, *E. coli* / *Salmonella* → moderate GC
- **High-GC organisms often show greater genome stability**
- **Although the 16S rRNA gene is conserved, GC content variation reflects evolutionary and taxonomic differences among bacterial genera**



Interpreting BLAST Identity

% Identity	Meaning
$\geq 97\%$	Same species
90–97%	Same genus
$< 90\%$	Distant

- What I observed : $\geq 99\%$ identity, Query coverage $\approx 100\%$ & E-value ≈ 0
- So, Sequence identity confirms: Correct Organism, correct gene and no contamination.
- Therefore, High percent identity and complete query coverage confirmed accurate species-level classification of the bacterial sequences.



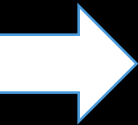
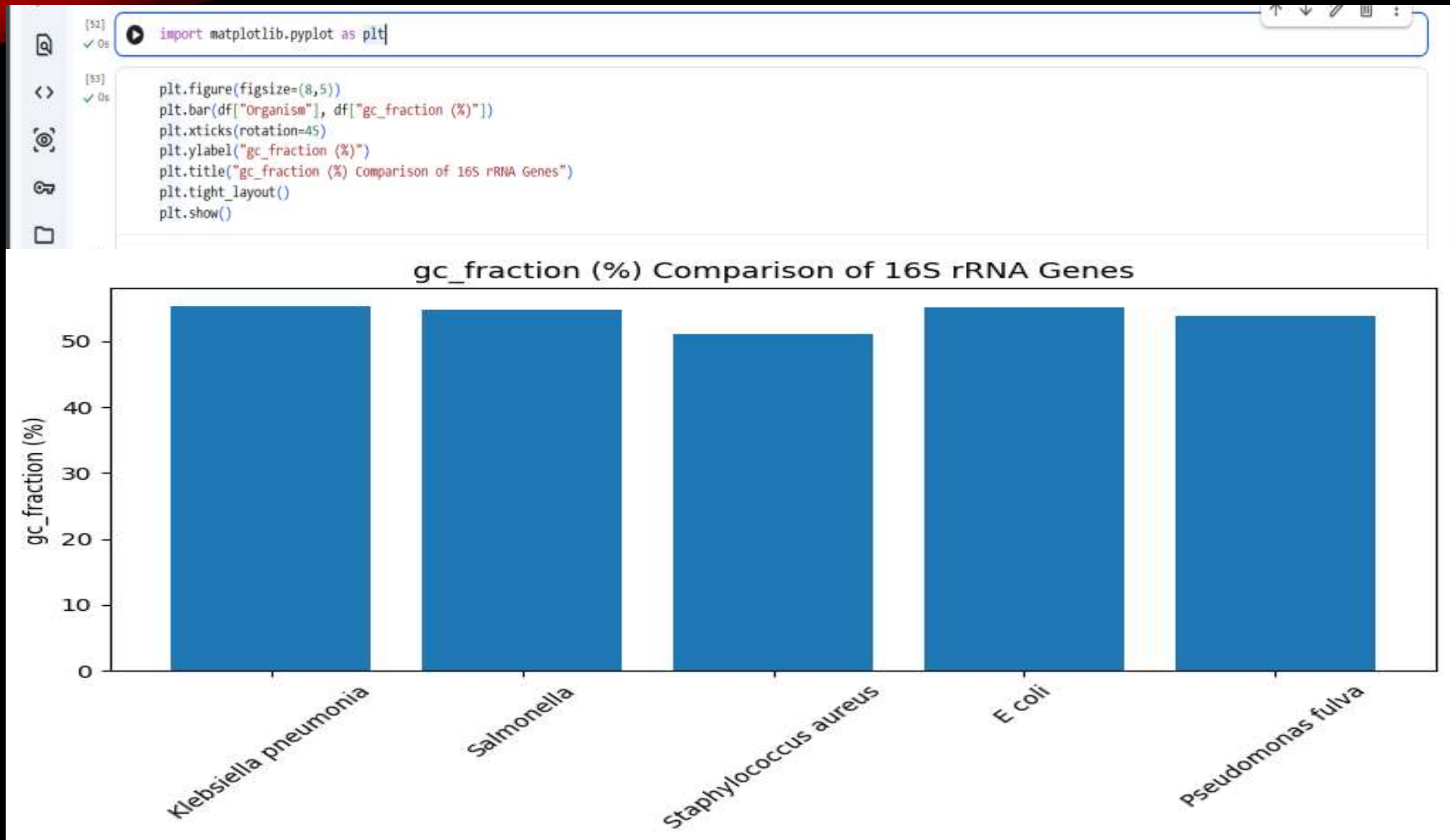
Interpreting E-value (Probability of match by chance)

E-value	Interpretation
0 / $<1e-10$	Highly significant
>0.01	Weak

- In my observation: E-value = 0 or near 0
- This signals about the fact that Near-zero E-values indicate statistically significant alignments, validating the reliability of BLAST matches
- Overall, I can conclude that using 16S rRNA gene sequences, I confirmed bacterial identities through BLAST analysis and observed conserved gene length with moderate GC variation, reflecting evolutionary relationships across species

F. VISUALIZED COMPARATIVE PATTERNS USING MATPLOTLIB

1)

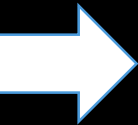
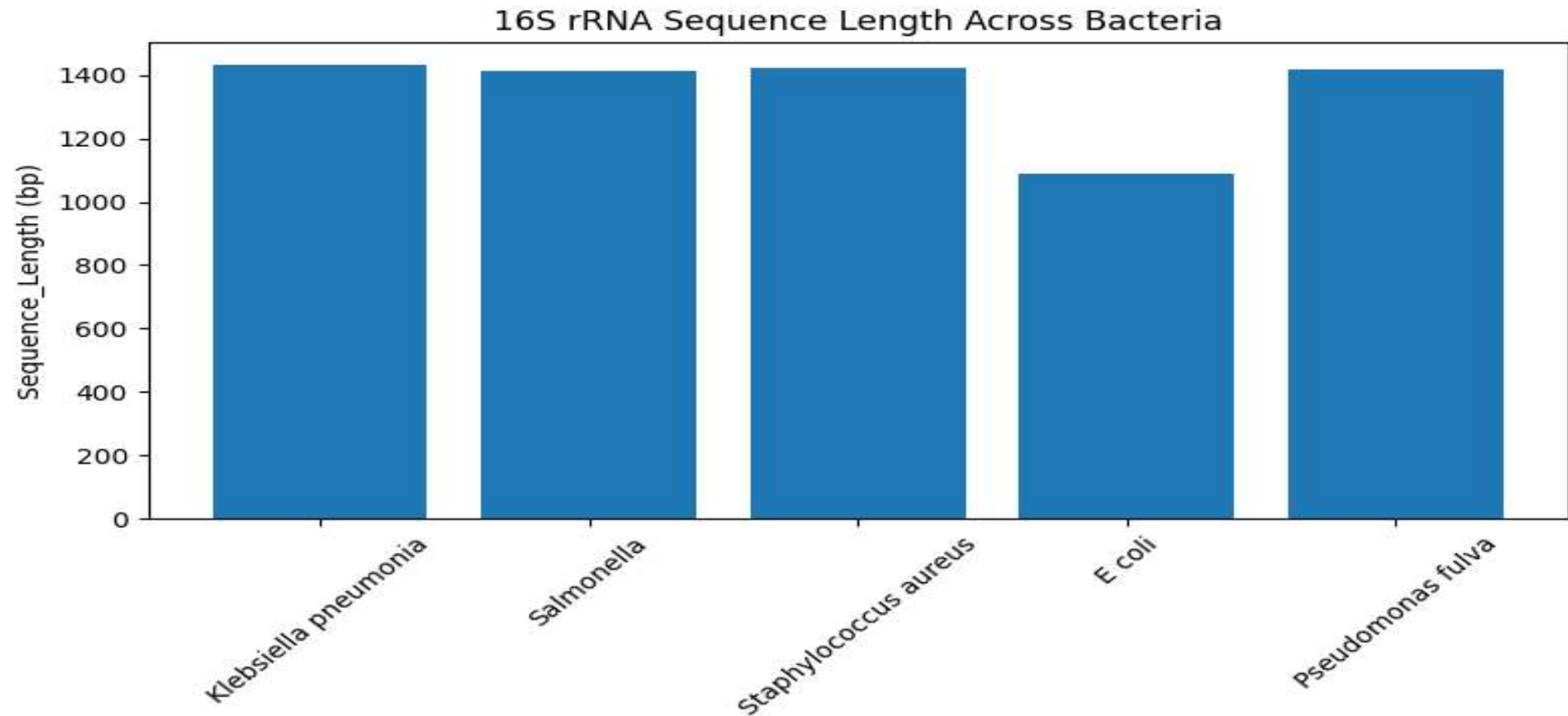


2)



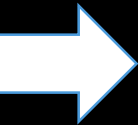
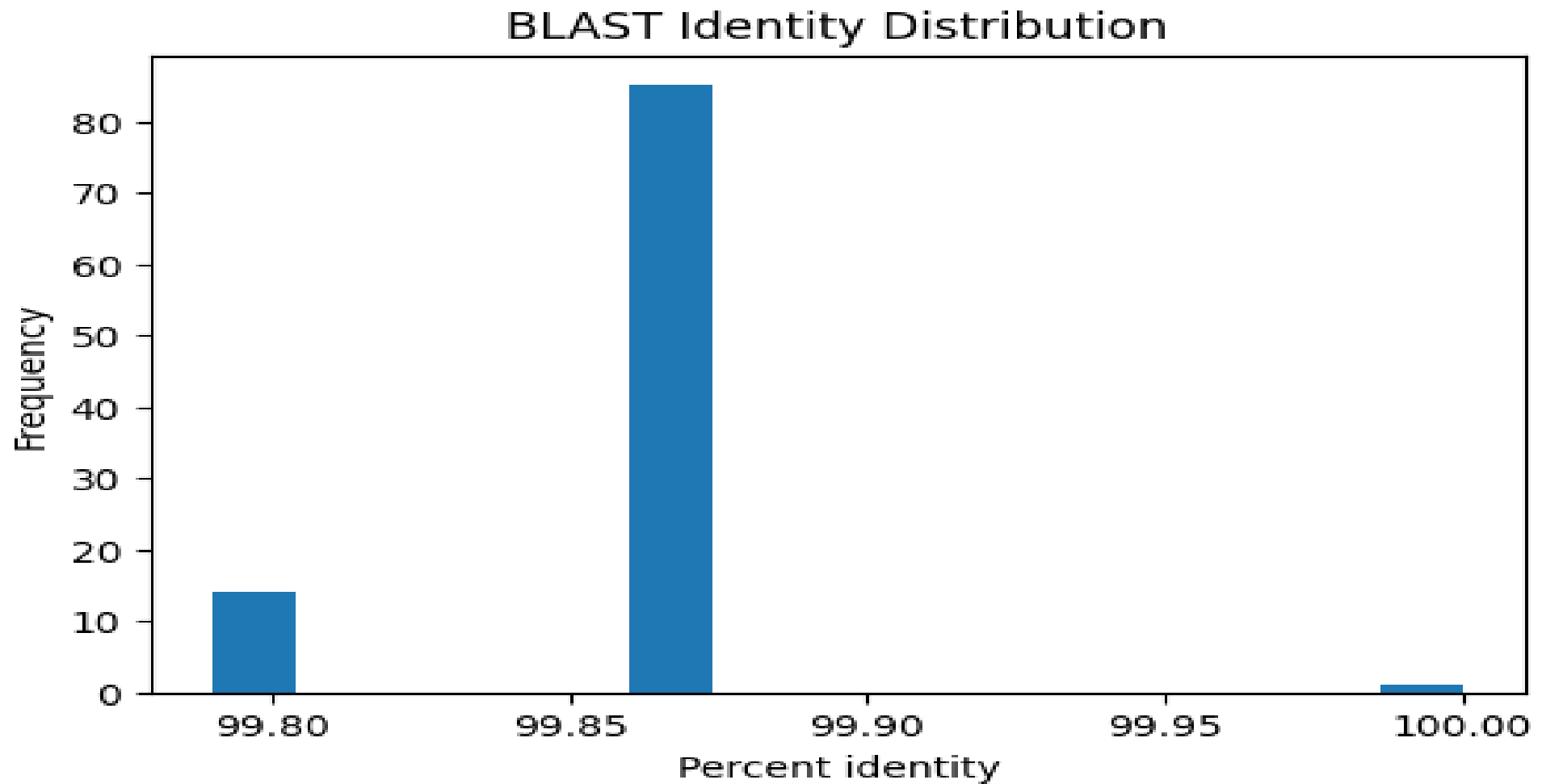
[59]
✓ 0s

```
plt.figure(figsize=(8,5))  
plt.bar(df["Organism"], df["Sequence_Length"])  
plt.xticks(rotation=45)  
plt.ylabel("Sequence_Length (bp)")  
plt.title("16S rRNA Sequence Length Across Bacteria")  
plt.tight_layout()  
plt.show()
```



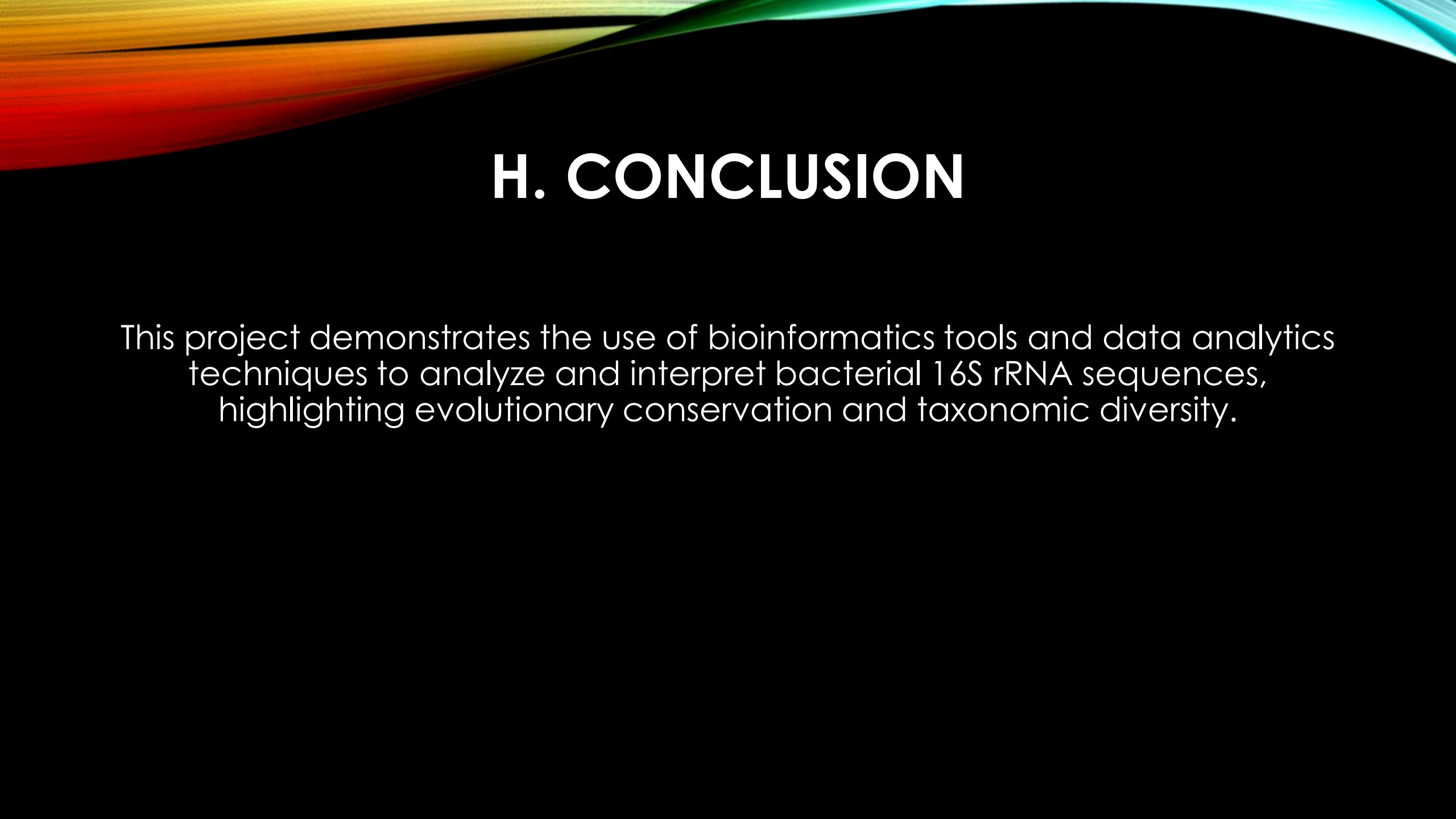
3)

```
[68] ✓ 0s  
plt.figure(figsize=(6,4))  
plt.hist(blast_clean["Percent identity"], bins=15)  
plt.xlabel("Percent identity")  
plt.ylabel("Frequency")  
plt.title("BLAST Identity Distribution")  
plt.show()
```



G. KEY RESULTS & INSIGHTS

- 16S rRNA gene lengths were highly conserved across species
- GC fraction showed moderate variation reflecting taxonomic differences
- BLAST analysis confirmed species identity with $\geq 99\%$ sequence identity
- Visualizations:
 1. GC Content comparison across bacterial species
 2. Sequence length comparison
 3. BLAST identity distribution plots



H. CONCLUSION

This project demonstrates the use of bioinformatics tools and data analytics techniques to analyze and interpret bacterial 16S rRNA sequences, highlighting evolutionary conservation and taxonomic diversity.